Gaussian 09 Tutorial

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Three key features of theory are required for *quantum chemical* calculations.

- Understand how initial specification of nuclear positions is used to calculate energy:
  - Solving the Schrödinger equation
- Understand "basis sets" and how to choose them:
  - Functions that represent the atomic orbitals
  - e.g., 3-21G, 6-311++G(3df,2pd), cc-pVTZ
- Understand levels of theory and how to choose them:
  - Wavefunction methods: Hartree-Fock, MP4, CI, CAS
  - Density functional methods: LYP, B3LYP, etc.
  - Compound methods: CBS, G3
  - Semiempirical methods: AM1, PM3
- Then, calculate properties.

General information of Gaussian 09

Part 1: Preparing Input Files

- **from Protein Data Bank (PDB) or databases**
- **Build a Structure using GaussView, Avogadro**

Part 2: Running Gaussian 09 on PC and HPC unit

Part 3: Visualization with GaussView

Part 4: Practice

General information of Gaussian 09

- Gaussian website

  www.gaussian.com

- Capability in Gaussian 09
  - ➢ Energy, Geometry optimization
    - ✓ DFT, TDDFT
  - ➢ Solvation models
  - ➢ Molecular properties
    - ✓ NMR, UV-vis, IRC, partial charges
    - ✓ Dipole moment, molecular orbitals
    - ✓ Electrostatic potential charge, Mulliken population

Preparing input files

- Running files
  - name.com, name.gjf : Gaussian input describing the desired calculation
  - name.sh : executable script
  - name.log : output file
  - fort.7 : molecular orbital file
- Scratch files
  - name.chk : check point file
  - name.rwk : read-write file
  - name.int : two electron integral file
  - name.d2e : two electron integral derivative file

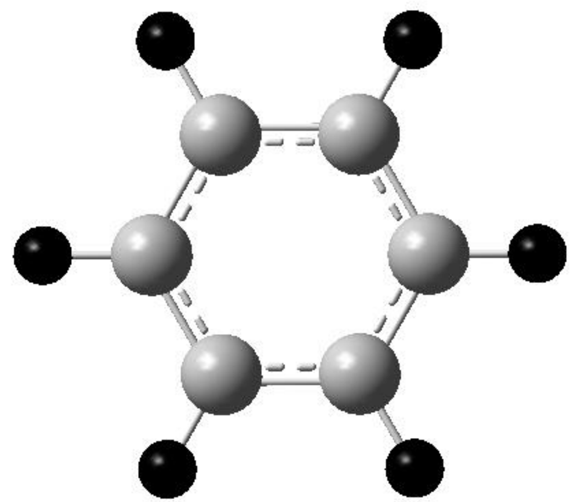
Preparing input files

- .com file in Gaussian 09
  - %chk = : path of the check point file
  - %mem = : specifying resource requirements
  - %nproc= : number of CPU

```
%chk=chk.chk
%mem=2GB
%nproc=2
# b3lyp/6-311g**

Title Card Required

0 1
 C                  -0.37656906    0.52301254    0.00000000
 C                   1.01859094    0.52301254    0.00000000
 C                   1.71612894    1.73076354    0.00000000
 C                   1.01847494    2.93927254   -0.00119900
 C                  -0.37635006    2.93919454   -0.00167800
 C                  -1.07395106    1.73098854   -0.00068200
 H                  -0.92632806   -0.42930446    0.00045000
 H                   1.56809894   -0.42950046    0.00131500
 H                   2.81580894    1.73084354    0.00063400
 H                   1.56867494    3.89141554   -0.00125800
 H                  -0.92647206    3.89147554   -0.00263100
 H                  -2.17355506    1.73117154   -0.00086200
```



Preparing input files

- .com or .gjf file in Gaussian 09
  - ➢ Calculation method
    - ✓ **Functional / basis set**
  - ➢ Calculation option

sp : single point energy
opt : geometry optimization
freq : frequency
irc : intrinsic reaction coordinate
pop : population
td : time dependent dft
scrf : self consistent reaction field

```
%chk=chk.chk
%mem=2GB
%nproc=2
# b3lyp/6-311g**

Title Card Required

0 1
 C                 -0.37656906    0.52301254    0.00000000
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 C                 -0.37635006    2.93919454   -0.00167800
 C                 -1.07395106    1.73098854   -0.00068200
 H                 -0.92632806   -0.42930446    0.00045000
 H                  1.56809894   -0.42950046    0.00131500
 H                  2.81580894    1.73084354    0.00063400
 H                  1.56867494    3.89141554   -0.00125800
 H                 -0.92647206    3.89147554   -0.00263100
 H                 -2.17355506    1.73117154   -0.00086200
```
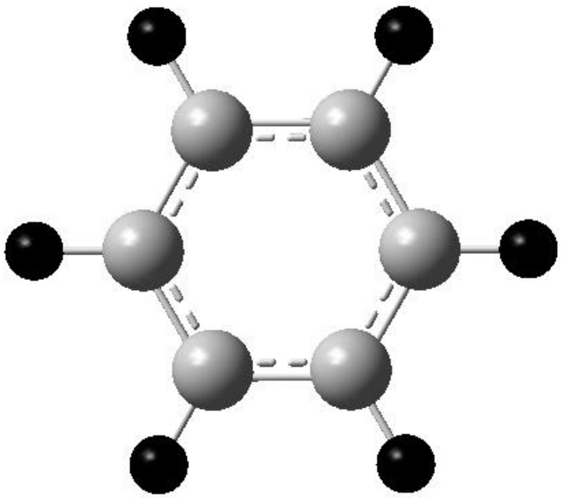
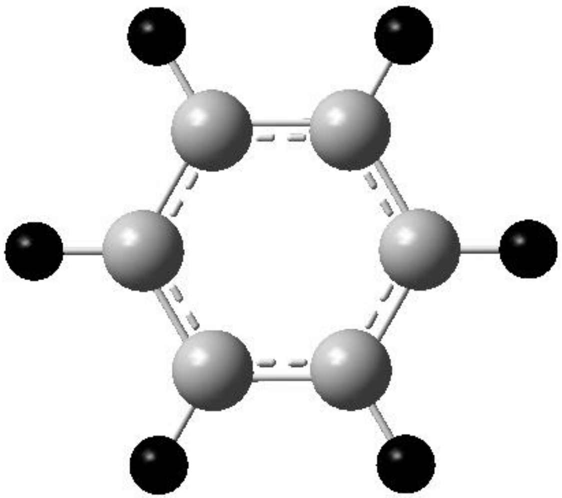



Preparing input files

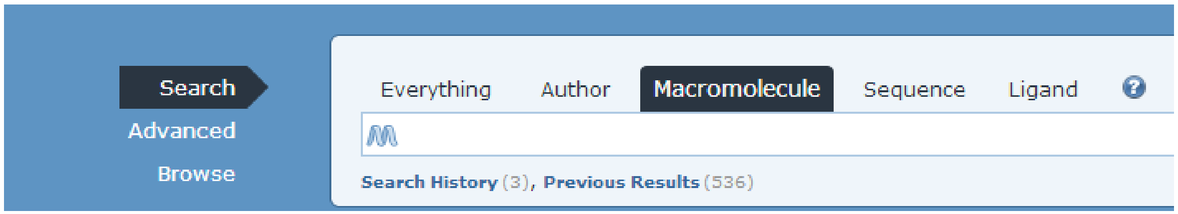
- Title
- Charge, spin multiplicity
- Molecule coordinate

```
%chk=chk.chk
%mem=2GB
%nproc=2
# b3lyp/6-311g**

Title Card Required

0 1
 C                 -0.37656906    0.52301254    0.00000000
 C                  1.01859094    0.52301254    0.00000000
 C                  1.71612894    1.73076354    0.00000000
 C                  1.01847494    2.93927254   -0.00119900
 C                 -0.37635006    2.93919454   -0.00167800
 C                 -1.07395106    1.73098854   -0.00068200
 H                 -0.92632806   -0.42930446    0.00045000
 H                  1.56809894   -0.42950046    0.00131500
 H                  2.81580894    1.73084354    0.00063400
 H                  1.56867494    3.89141554   -0.00125800
 H                 -0.92647206    3.89147554   -0.00263100
 H                 -2.17355506    1.73117154   -0.00086200
```



Preparing input files (PDB)

PDB-101




```
HETATM    1  C           0      -0.619   0.612   0.000              C
HETATM    2  C           0       0.776   0.612   0.000              C
HETATM    3  C           0       1.474   1.820   0.000              C
HETATM    4  C           0       0.776   3.028  -0.001              C
HETATM    5  C           0      -0.619   3.028  -0.002              C
HETATM    6  C           0      -1.316   1.820  -0.001              C
HETATM    7  H           0      -1.169  -0.341   0.000              H
HETATM    8  H           0       1.326  -0.341   0.001              H
HETATM    9  H           0       2.574   1.820   0.001              H
HETATM   10  H           0       1.326   3.980  -0.001              H
HETATM   11  H           0      -1.169   3.980  -0.003              H
HETATM   12  H           0      -2.416   1.820  -0.001              H
```

benzene.pdb

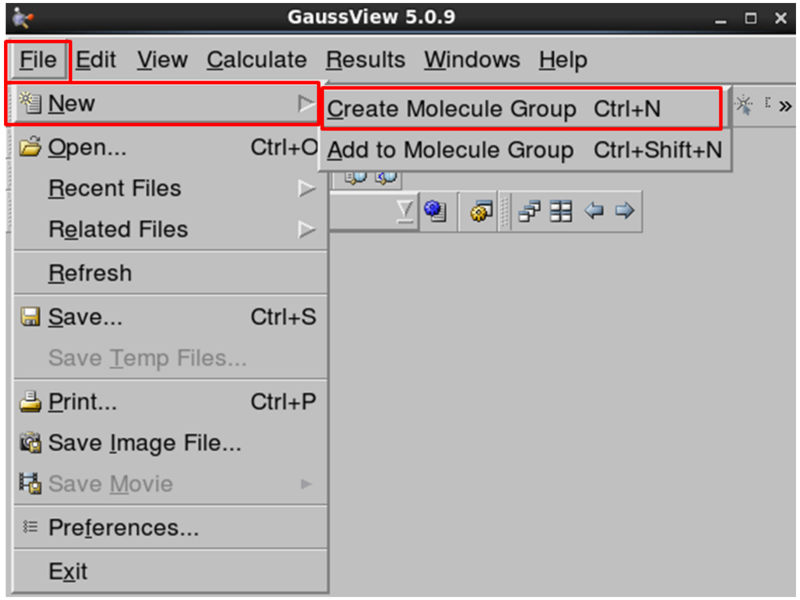
HETATM	1	C	0	-0.619	0.612	0.000	C
HETATM	2	C	0	0.776	0.612	0.000	C
HETATM	3	C	0	1.474	1.820	0.000	C
HETATM	4	C	0	0.776	3.028	-0.001	C
HETATM	5	C	0	-0.619	3.028	-0.002	C
HETATM	6	C	0	-1.316	1.820	-0.001	C
HETATM	7	H	0	-1.169	-0.341	0.000	H
HETATM	8	H	0	1.326	-0.341	0.001	H
HETATM	9	H	0	2.574	1.820	0.001	H
HETATM	10	H	0	1.326	3.980	-0.001	H
HETATM	11	H	0	-1.169	3.980	-0.003	H
HETATM	12	H	0	-2.416	1.820	-0.001	H

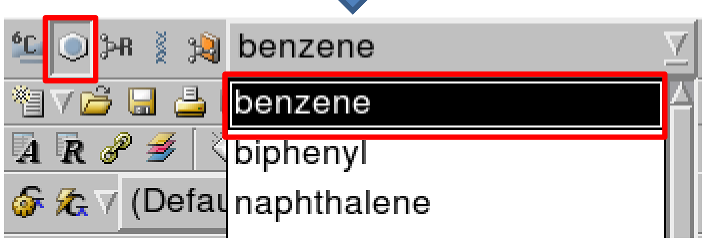
pdb2xyz

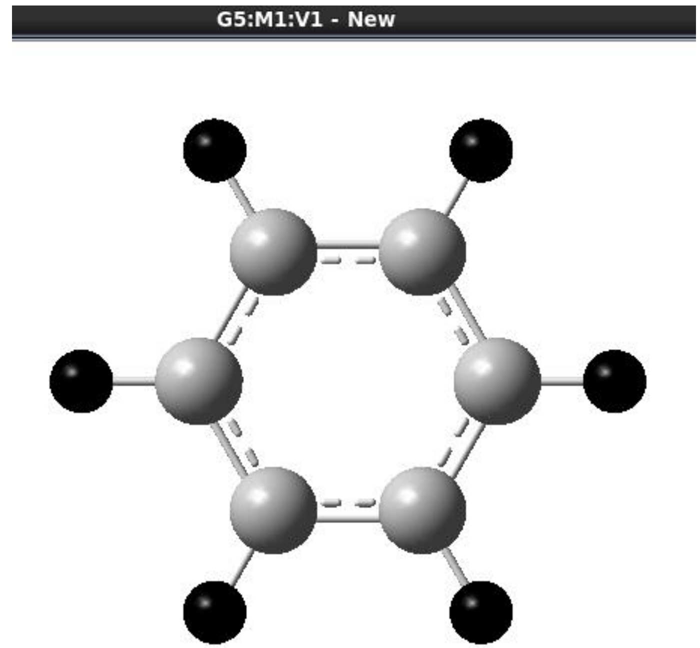
C	0.00000000	0.00000000	0.00000000
C	1.39500000	0.00000000	0.00000000
C	2.09300000	1.20800000	0.00000000
C	1.39500000	2.41600000	-0.00100000
C	0.00000000	2.41600000	-0.00200000
C	-0.69700000	1.20800000	-0.00100000
H	-0.55000000	-0.95300000	0.00000000
H	1.94500000	-0.95300000	0.00100000
H	3.19300000	1.20800000	0.00100000
H	1.94500000	3.36800000	-0.00100000
H	-0.55000000	3.36800000	-0.00300000
H	-1.79700000	1.20800000	-0.00100000

http://www.rcsb.org/pdb/home/home.do

Preparing input files (GaussView)





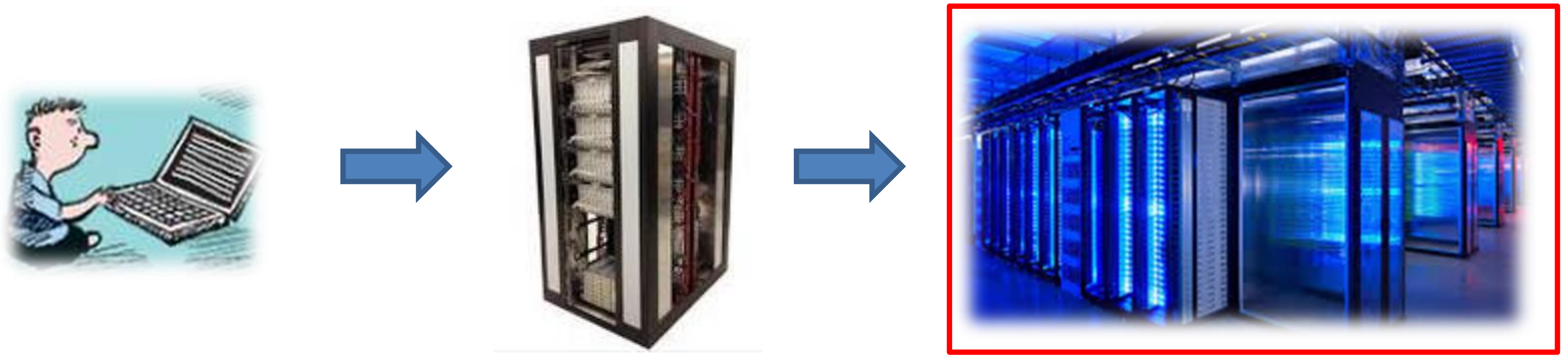


Preparing input files

```
#!/bin/bash
# @ job_type =serial
# @ class = large
# @ error = f2log.err
# @ output = f2log.out
# @ notification = complete
# @ notify_user =
# @ resources = ConsumableCpus(1) ConsumableMemory(4gb)
# @ cluster_list = tu.tusmp
# @ wall_clock_limit=72:00:00
# @ queue

export MEMORY_AFFINITY=MCM
export OMP_NUM_THREADS=1
export g03root="/applic/Applications/G09"
export GAUSS_SCRDIR="/pwork01/$USER/G09_SCR"
mkdir -p $GAUSS_SCRDIR
. $g03root/g09/bsd/g09.profile
/applic/Applications/G09/g09/g09 01.com > 01.log
```

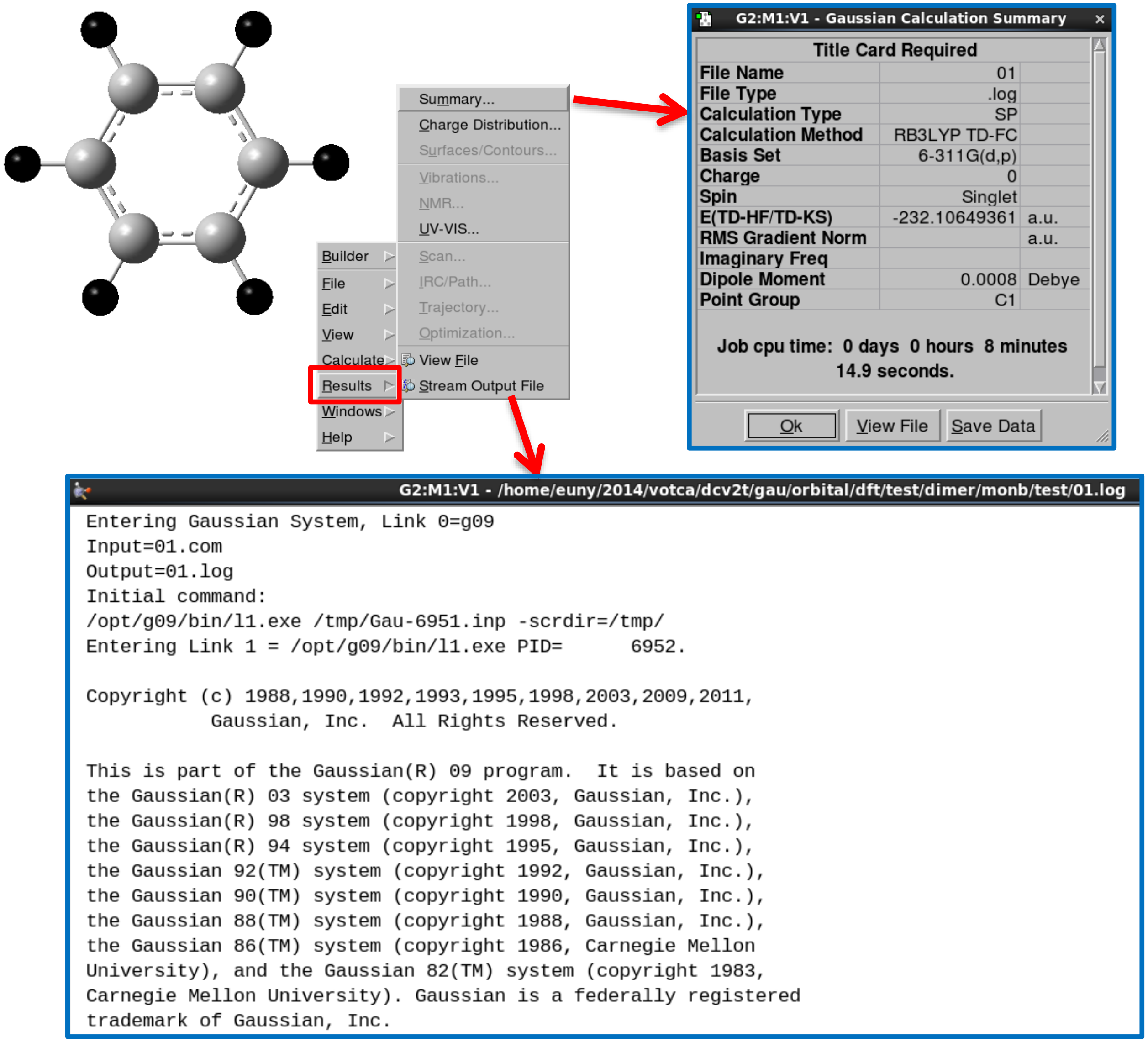
- User email
- Number of CPU, Memory
- Calculation cluster
- Job time
- Job file name

Running Gaussian 09 on supercomputer

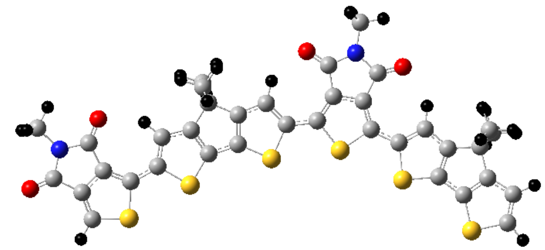


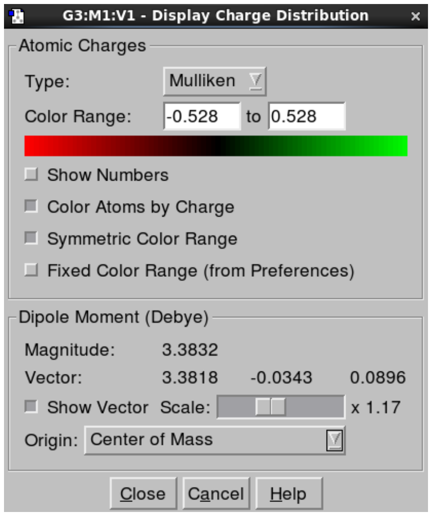
- Login node > calculation node
  - ssh account
- Keywords
  - qstate –q #system status
  - qsub test.pbs or batch file #job running

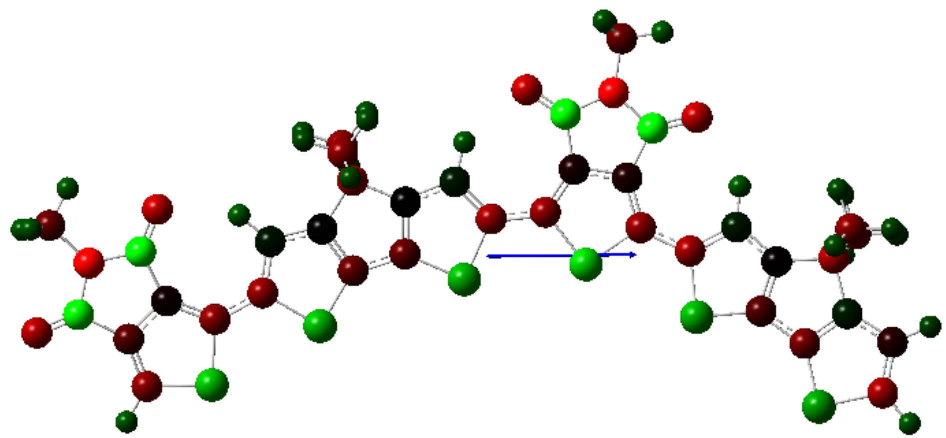
Visualization with GaussView

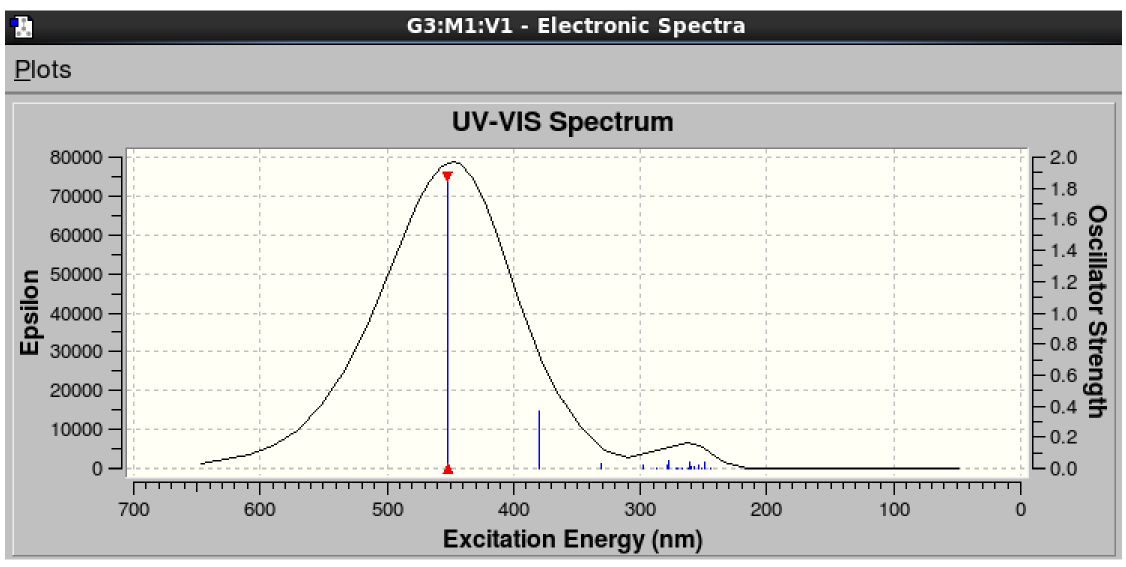


Visualization with GaussView

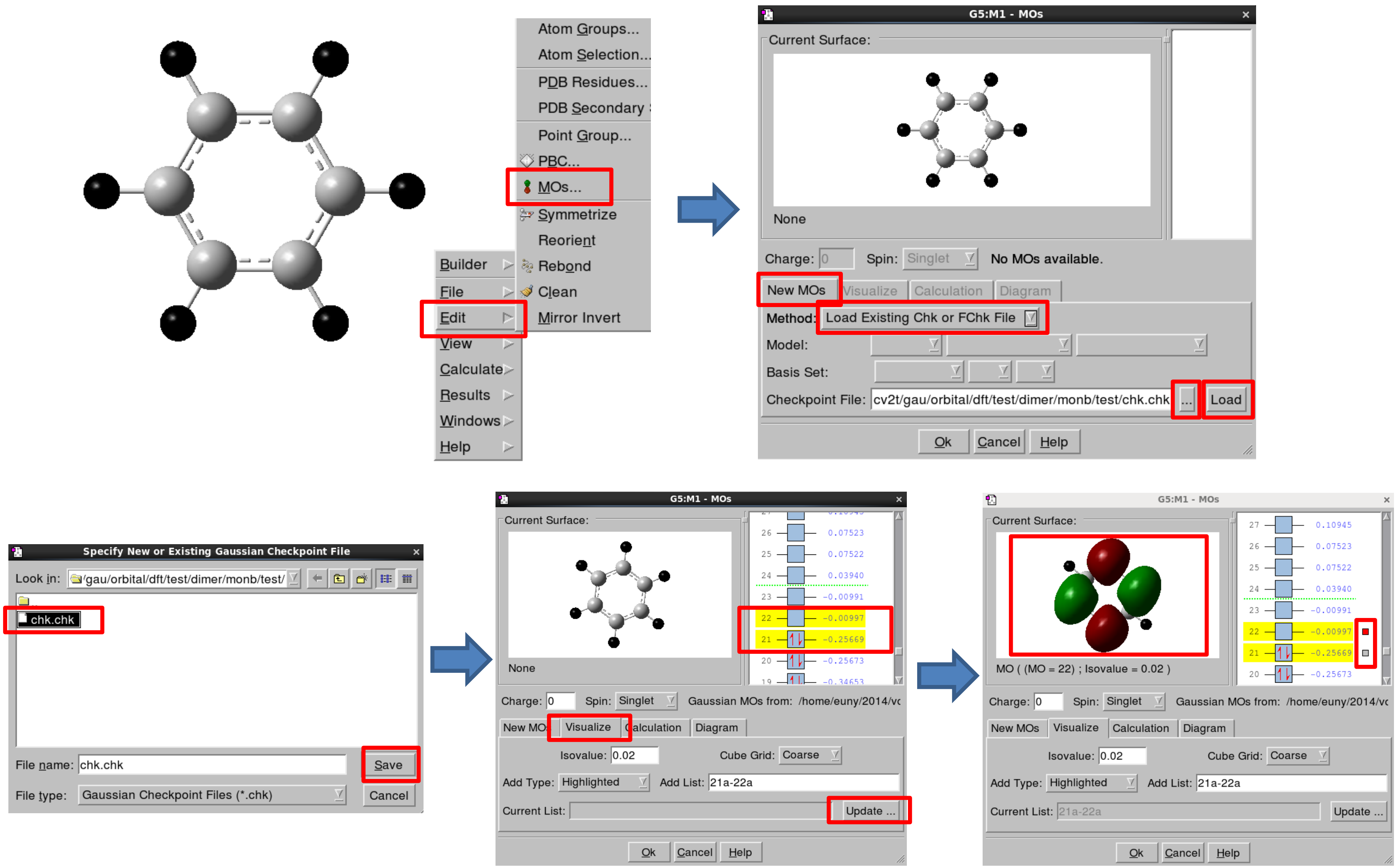








Visualization with GaussView



Practice

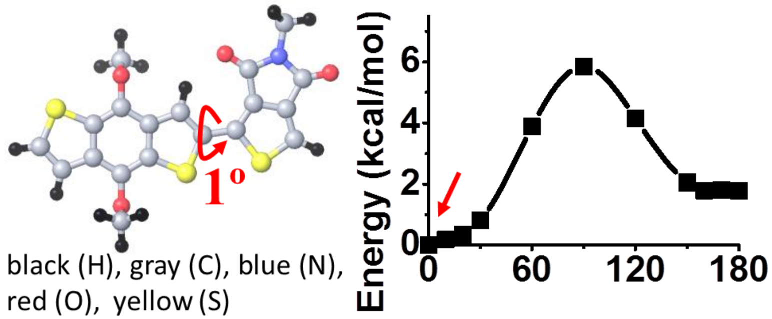
1. Geometry optimization

2. Molecular orbital

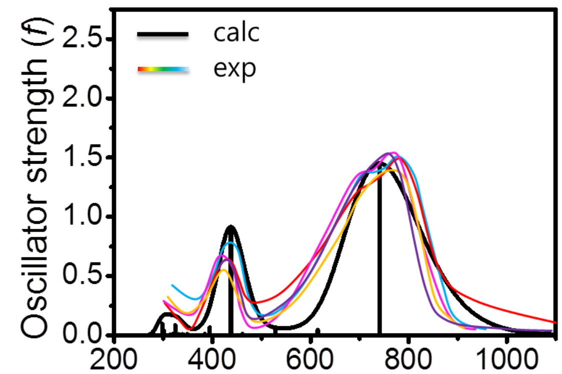
3. Time dependent DFT method

4. UV-vis spectrum

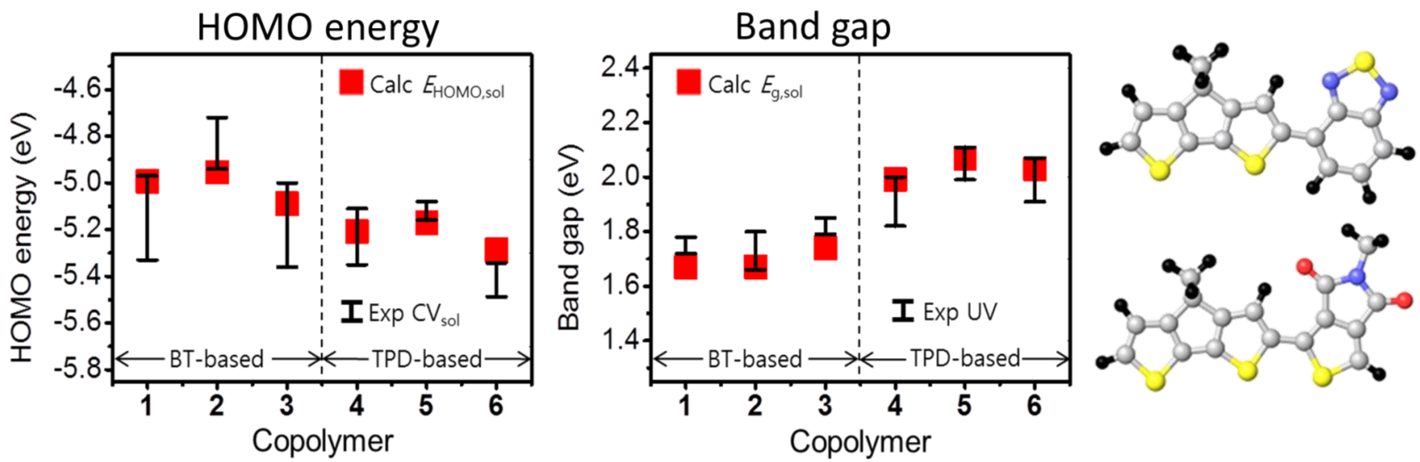
- Torsion energy curves



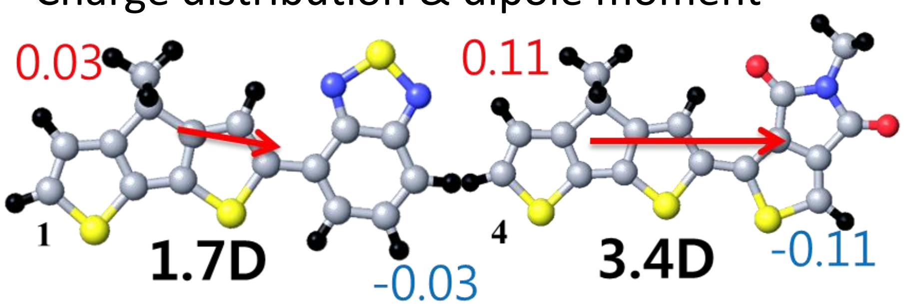
- Absorption spectra



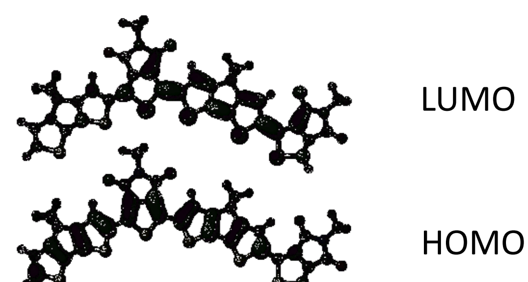
- Electronic structure



- Charge distribution & dipole moment



- Molecular orbital



Practice files

- Optimization

```
%chk=chk.chk
%mem=4GB
%nproc=1
# opt=(maxcycle=999) scf=(maxcycle=999) b3lyp/6-311g**

Benzene Optimization

0 1
 C                  -0.37656906    0.52301254    0.00000000
 C                   1.01859094    0.52301254    0.00000000
 C                   1.71612894    1.73076354    0.00000000
 C                   1.01847494    2.93927254   -0.00119900
 C                  -0.37635006    2.93919454   -0.00167800
 C                  -1.07395106    1.73098854   -0.00068200
 H                  -0.92632806   -0.42930446    0.00045000
 H                   1.56809894   -0.42950046    0.00131500
 H                   2.81580894    1.73084354    0.00063400
 H                   1.56867494    3.89141554   -0.00125800
 H                  -0.92647206    3.89147554   -0.00263100
 H                  -2.17355506    1.73117154   -0.00086200
```

Practice files

- Optimization

```
%chk=chk.chk
%mem=4GB
%nproc=1
# opt=(maxcycle=999) scf=(maxcycle=999) pop=minimal b3lyp/6-311g**

Benzene Molecular Orbital

0 1
 C                  -0.37656906    0.52301254    0.00000000
 C                   1.01859094    0.52301254    0.00000000
 C                   1.71612894    1.73076354    0.00000000
 C                   1.01847494    2.93927254   -0.00119900
 C                  -0.37635006    2.93919454   -0.00167800
 C                  -1.07395106    1.73098854   -0.00068200
 H                  -0.92632806   -0.42930446    0.00045000
 H                   1.56809894   -0.42950046    0.00131500
 H                   2.81580894    1.73084354    0.00063400
 H                   1.56867494    3.89141554   -0.00125800
 H                  -0.92647206    3.89147554   -0.00263100
 H                  -2.17355506    1.73117154   -0.00086200
```

Practice files

- TDDFT UV-Visible input

```
%chk=chk.chk
%mem=4GB
%nproc=1
# td=(singlets,Nstates=20) b3lyp/6-311g**

Benzene TDDFT

0 1
 C                  -0.37656906    0.52301254    0.00000000
 C                   1.01859094    0.52301254    0.00000000
 C                   1.71612894    1.73076354    0.00000000
 C                   1.01847494    2.93927254   -0.00119900
 C                  -0.37635006    2.93919454   -0.00167800
 C                  -1.07395106    1.73098854   -0.00068200
 H                  -0.92632806   -0.42930446    0.00045000
 H                   1.56809894   -0.42950046    0.00131500
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 H                  -0.92647206    3.89147554   -0.00263100
 H                  -2.17355506    1.73117154   -0.00086200
```

Hands-on (GV6)

Basics concept QM HY=EY
Numerical solution PDE

GV Visualuizer
G09 calculation

Opt freq calculation for ground state chemistry

opt /freq bottle neck

Opt freq calculation for transition state chemistry

excited state uv-visible spectra

oniom algorithm QM/MM for large molecules

Exploring the PES with Gaussian/GaussView

Gaussian Input File

example input for water

```
%chk=h2o_scan.chk
%mem=6MW
%nproc=1
# hf/3-21g opt

Title

0 1
O    0.0   0.0   0.0
H    1.0   0.0   0.0
H    0.0   1.0   0.0
```

link 0 commands

route line

charge and multiplicity

geometry in cartesian coordinates

Geometry Specification

2 commons ways to specify the molecular geometry

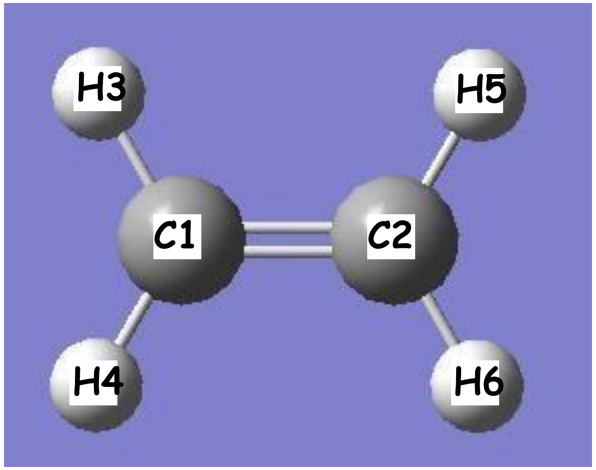
1. Cartesian coordinates:
   - atomic symbol, x, y, z coordinates of each nucleus
   - Gaussian expects values in Angstroms
   - convenient because most molecular building programs will output Cartesian coordinates

2. Z-matrix coordinates:
   - also called internal coordinates
   - specify positions of atoms relative to one another using bond lengths, angles and dihedral angles (3N-6 variables)
   - one section specifies connectivity, second section specifies values of variables corresponding to bond lengths, etc.
   - Gaussian expects values in Angstroms and degrees
   - convenient for PES scans because bonds and angles are defined explicitly

Z-matrix Input

connectivity specification:

```
C
C   1   B1
H   1   B2   2  A1
H   1   B3   2  A2   3  D1
H   2   B4   1  A3   3  D2
H   2   B5   1  A4   5  D3
```

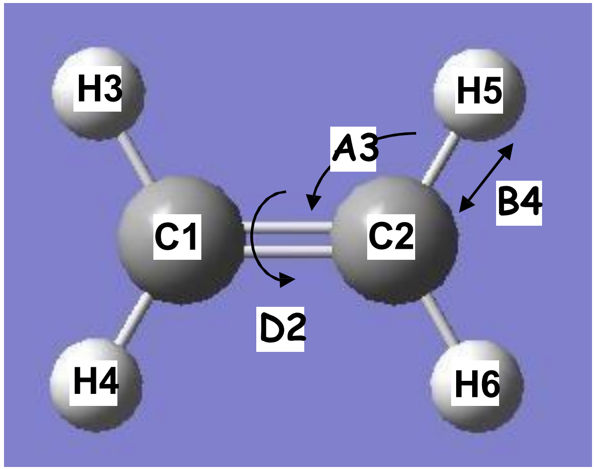


- 1st column specifies atom type
- 2nd column defines a bond, e.g. the „1“ in line 2 indicates that atom 2 is bonded to atom 1
- 3rd column gives the label of a variable corresponding to the bond length
- 4th column defines an angle, e.g. the „2“ in line 3 indicates that the 3rd atom forms a 3-1-2 (H3-C1-C2) angle
- 5th line gives the label of a variable containing the value of the dihedral angle
- 6th line defines a dihedral angle, e.g. the „3“ in line 4 indicates that the 4th atom forms a dihedral 4-1-2-3 (H4-C1-C2-H3) dihedral angle
- 7th line gives the label of a variable containing the value of the dihedral angle

Z-matrix Input

connectivity specification:

```
C
C   1   B1
H   1   B2   2  A1
H   1   B3   2  A2   3  D1
H   2   B4   1  A3   3  D2
H   2   B5   1  A4   5  D3
```



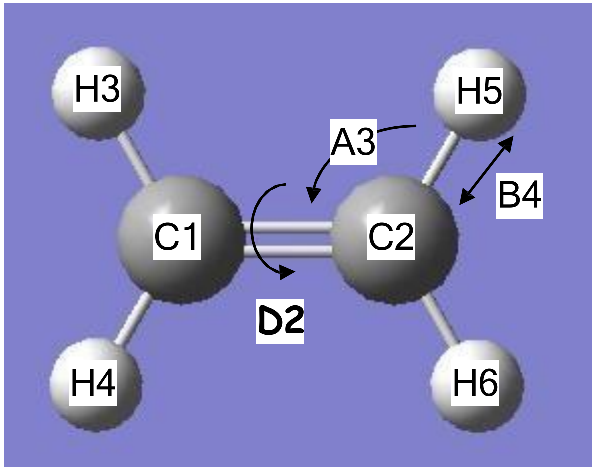
example:

- line 5 means: a hydrogen atom is bonded atom 2 with a bond distance of B4, forms an angle with atoms 2 and 1 with a value of A3, and forms a dihedral angle with atoms 2, 1, and 3 with a value of D2

Z-matrix Input

connectivity specification:

```
C
C  1  B1
H  1  B2  2  A1
H  1  B3  2  A2  3  D1
H  2  B4  1  A3  3  D2
H  2  B5  1  A4  5  D3
```



variables:

B1=1.5
B2=1.1
B3=1.1
B4=1.1
B5=1.1
A1=120.0
A2=120.0
A3=120.0
A4=120.0
D1=0.0
D2=0.0
D3=180.0

- can simplify by taking advantage of symmetry
  - expect C-H bonds to be same lengths
    → use variable B2 for all C-H bonds
  - expect H-C-C angles to be the same
    → use variable A1 for all H-C-C angles

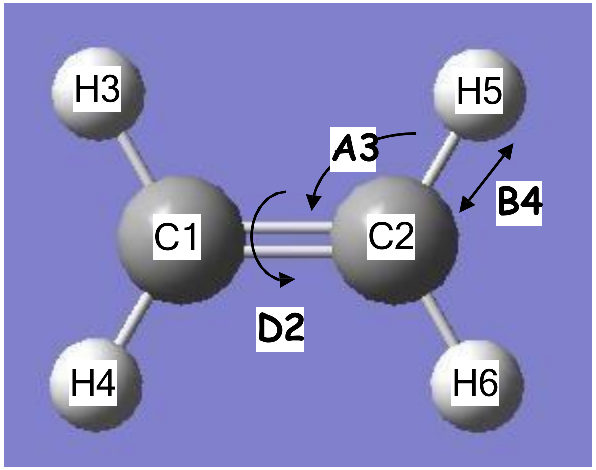
Z-matrix Input

connectivity specification:

```
C
C  1  B1
H  1  B2  2  A1
H  1  B2  2  A1  3  D1
H  2  B2  1  A1  3  D2
H  2  B2  1  A1  5  D3
```

variables:

B1=1.5
B2=1.1
A1=120.0
D1=0.0
D2=0.0
D3=180.0



- can simplify by taking advantage of symmetry
  - expect C-H bonds to be same lengths
    $\rightarrow$ use variable B2 for all C-H bonds
  - expect H-C-C angles to be the same
    $\rightarrow$ use variable A1 for all H-C-C angles
- careful, though
  - assigning the same label to two or more geometric variables means they have to remain equal throughout entire calculation

Wavefunction Expansion

Job Type method/basis_set keyword1=(options) keyword2=(options)

- sp : single point energy
- opt : geometry optimization
- freq : frequency
- irc : intrinsic reaction coordinate
- pop : population
- td : time dependent dft
- scrf : self consistent reaction field

- HF
- SCF
- CASSCF
- DFT/ B3lyp
- CAM-B3lyp
- TDDFT
- CISD

Route Line

- specifies type of calculation that is to be performed
- line starts with „#", can only be 80 characters in length
- can have multiple lines
- line contains method, basis set and keywords with options in parentheses:

  # method/basis_set keyword1=(options) keyword2=(options)

  # keyword3(options) keyword4=(options)

- must be followed by a blank line
- full list of keywords and options available at:

  http://www.gaussian.com/g_ur/keywords.htm

- relevant keywords for exploring the PES:
- scan → perform a scan along predefined coordinates
- opt → perform a geometry optimization to a minimum or transition state
- freq → perform a frequency/normal mode calculation

Output

- gaussian output files will usually end with .log or .out
- contains a lot of information → contents depend on type of calculation
- units are usually Hartree for energy and Angstrom for distance (but not always)

1 Hartree = 627.51 kcal/mol

1 Angstrom = 1.0 x 10^{-10} m

Things to look for in the output:

- molecular structure → look for a line saying “Input orientation:”
- molecular energy → look for a line saying “SCF Done:”
- convergence in optimization → look for a line saying “Maximum Force”
- summary of a rigid scan → look for a line saying “Summary of the potential surface scan”
- summary of a relaxed scan → look for a line saying “Summary of Optimized Potential Surface Scan”
- frequency information → look for a line saying “Harmonic frequencies”

Example Calculations

1. Single point calculation of ethane
2. Rigid scan of the C-C bond in ethane
3. Geometry optimization of $(CH_3)_2CO$
4. Transition state search for $(CH_3)_2CO \rightarrow CH_3C(OH)CH_2$
5. Relaxed scan of the O-H bond for $(CH_3)_2CO \rightarrow CH_3C(OH)CH_2$
6. Frequency calculation of $(CH_3)_2CO$

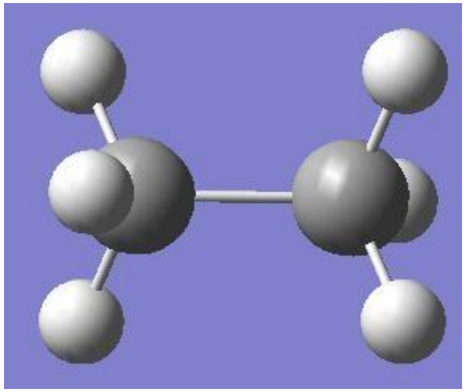
Single Point Calculation

Single Point Calculation:

- calculate the energy for a specific geometry
- provides 1 point on the potential energy surface
- the geometry is not updated or changed
- simplest, yet most fundamental, type of calculation

This example:

- calculate the energy of ethane
- geometry provided in Z-matrix format
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Single-Point Energy Calculation - Input

```
# hf/3-21g

Title

0 1
 C
 H    1   B1
 H    1   B2   2   A1
 H    1   B3   2   A2   3   D1
 C    1   B4   4   A3   2   D2
 H    5   B5   1   A4   4   D3
 H    5   B6   1   A5   4   D4
 H    5   B7   1   A6   4   D5

   B1         1.07
   B2         1.07
   B3         1.07
   B4         1.53
   B5         1.07
   B6         1.07
   B7         1.07
   A1       109.5
   A2       109.5
   A3       109.5
   A4       109.5
   A5       109.5
   A6       109.5
   D1      -120.0
   D2       120.0
   D3        60.0
   D4       180.0
   D5       -60.0
```

- route line specifies method used in calculation
- coordinates of ethane in Z-matrix format

Single-Point energy Calculation - Output

in a single point calculation we may want to determine:

- the energy of the given structure

```
SCF Done:  E(RHF) =  -78.3232725661     A.U. after    9 cycles
            Convg  =     0.7799D-08              -V/T =  1.9647
            S**2   =    0.0000
```

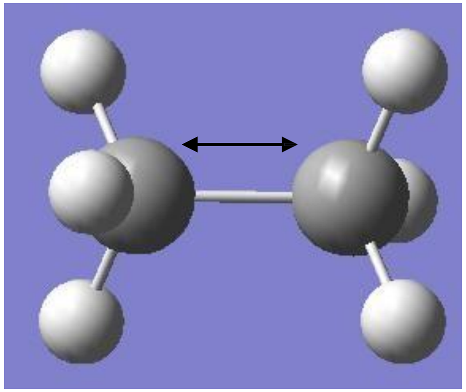
- other properties of the system
  - we"ll look into those more in future lectures

Rigid Scan Calculation:

- calculate the energies for a series of structures
- structures are based on predetermined changes to an initial structure, e.g. varying a bond length or changing an angle
- all non-scanned geometric variables are fixed at their original values
- provides a series of points on the potential energy surface

This example:

- perform a rigid scan by increasing the C-C bond length in ethane
- geometry provided in Z-matrix format $\rightarrow$ required by gaussian to do a scan
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Rigid Scan - Input

```
# scan hf/3-21g nosymm

Title

0 1
 C
 H    1   B1
 H    1   B2   2   A1
 H    1   B3   2   A2   3   D1
 C    1   B4   4   A3   2   D2
 H    5   B5   1   A4   4   D3
 H    5   B6   1   A5   4   D4
 H    5   B7   1   A6   4   D5

   B1    1.070000
   B2    1.070000
   B3    1.070000
   B4    1.000000   60 0.1
   B5    1.070000
   B6    1.070000
   B7    1.070000
   A1    109.4712
   A2    109.4712
   A3    109.4712
   A4    109.4712
   A5    109.4712
   A6    109.4712
   D1   -120.0000
   D2    120.0000
   D3     60.0000
   D4    180.0000
   D5    -60.0000
```

- keyword „scan" request a rigid scan of the selected variables
- „hf/3-21G" specifies level of theory for the calculation
- „nosymm" tells the program to set the initial symmetry to C1
  - scans often change the symmetry of the system, causing the calculation to fail

- „B4 1.000000 60 0.1" tells the program to increase the value of variable B4 from an initial value of 1.0 Ang in 60 steps of 0.1 Ang
- this will result in 61 single point calculations with B4 ranging from 1.0 to 7.0 Ang in increments of 0.1 Ang
- all other variables will remain fixed at their original values
- more than one variable can be selected for scanning in a given input
- if two or more variables are scanned, the energy will be calculated for all possible combinations of the scanned variables

Rigid Scan - Output

in a rigid scan calculation we may want to determine:

- a series of energies as a function of the changed geometric variable

Summary of the potential surface scan:

N	B4	SCF
1	1.0000	-78.32327
2	1.1000	-78.54303
3	1.2000	-78.67097
4	1.3000	-78.74176
5	1.4000	-78.77721
6	1.5000	-78.79108
7	1.6000	-78.79176
8	1.7000	-78.78429
9	1.8000	-78.77168
10	1.9000	-78.75587
11	2.0000	-78.73819
12	2.1000	-78.71959
13	2.2000	-78.70077
14	2.3000	-78.68218
15	2.4000	-78.66416
16	2.5000	-78.64690
17	2.6000	-78.63050
18	2.7000	-78.61502
19	2.8000	-78.60044
20	2.9000	-78.58674
.	.	.
.	.	.
.	.	.

- value of the coordinate that was scanned

- energy in Hartree at each step of the scan

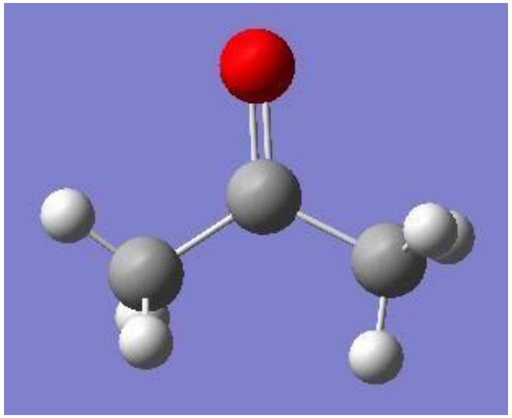
Note: if multiple coordinates were scanned the summary will contain multiple columns for those coordinates

Geometry Optimization:

- minimize the energy of a molecule by iteratively modifying its structure
- provides the energetically-preferred structure of a molecule
- the located structure will correspond to the local minimum nearest on the potential energy surface to the input structure
- suitable for determining the structures and energies of reactants and products

This example:

- optimize the geometry of $(CH_3)_2CO$ starting from a structure built with gaussview using standard bond lengths and angles
- geometry provided in Z-matrix format
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Geometry Optimization - Input

```
# opt hf/3-21g nosymm

Title Card Required

0 1
 C
 H    1  B1
 H    1  B2   2   A1
 H    1  B3   3   A2   2   D1
 C    1  B4   4   A3   3   D2
 O    5  B5   1   A4   4   D3
 C    5  B6   1   A5   6   D4
 H    7  B7   5   A6   1   D5
 H    7  B8   5   A7   1   D6
 H    7  B9   5   A8   1   D7

B1       1.07000000
B2       1.07000000
B3       1.07000000
B4       1.54000000
B5       1.43000000
B6       1.54000000
B7       1.07000000
B8       1.07000000
B9       1.07000000
A1     109.47120255
A2     109.47121829
A3     109.47121829
A4     120.00000000
A5     120.00000000
A6     109.47122063
A7     109.47122063
A8     109.47122063
D1     120.00000060
D2    -120.00003407
D3     180.00000000
D4    -180.00000000
D5     166.53624326
D6      46.53624326
D7     -73.46375674
```

- keyword „opt“ requests a geometry optimization to a minimum energy structure
- „hf/3-21G“ specifies level of theory for the calculation
- „nosymm“ tells the program to set the initial symmetry to C1
  - geometry optimizations sometimes change the symmetry of the system, causing the calculation to fail
  - the minimum energy structure may not have the same symmetry as the initial structure

Geometry Optimization - Output

in a geometry optimization we may want to determine:

- a stationary point corresponding to a minimum energy structure

  $\rightarrow$ need to monitor whether the convergence criteria are met

first step:

Item	Value	Threshold	Converged?
Maximum Force	0.196220	0.000450	NO
RMS Force	0.033395	0.000300	NO
Maximum Displacement	0.253014	0.001800	NO
RMS Displacement	0.052569	0.001200	NO

intermediate step:

Item	Value	Threshold	Converged?
Maximum Force	0.001162	0.000450	NO
RMS Force	0.000252	0.000300	YES
Maximum Displacement	0.260013	0.001800	NO
RMS Displacement	0.096764	0.001200	NO

final step:

Item	Value	Threshold	Converged?
Maximum Force	0.000058	0.000450	YES
RMS Force	0.000018	0.000300	YES
Maximum Displacement	0.001554	0.001800	YES
RMS Displacement	0.000615	0.001200	YES

Geometry Optimization - Output

in a geometry optimization we may want to determine:

- a stationary point corresponding to a minimum energy structure
  $\rightarrow$ need to monitor whether the convergence criteria are met
- energy of the optimized structure
  $\rightarrow$ energy statement in step where convergence criteria are met
  $\rightarrow$ last energy statement in the output file

```
SCF Done:  E(RHF) =  -190.887221224     A.U. after    9 cycles
            Convg  =    0.5730D-08             -V/T =  2.0019
            S**2   =   0.0000
```

Geometry Optimization - Output

in a geometry optimization we may want to determine:

- a stationary point corresponding to a minimum energy structure
  - $\rightarrow$ need to monitor whether the convergence criteria are met
- energy of the optimized structure
  - $\rightarrow$ energy statement in step where convergence criteria are met
  - $\rightarrow$ last energy statement in the output file
- geometry of the optimized structure
  - $\rightarrow$ follows statement where convergence criteria are met

Input orientation:

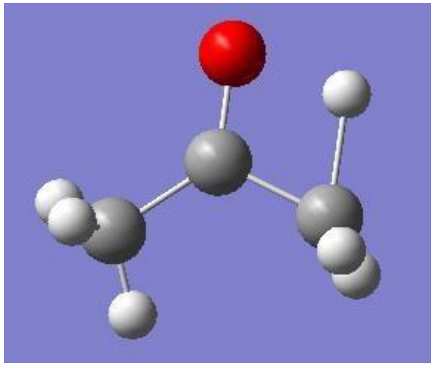
Center Number	Atomic Number	Atomic Type	Coordinates (Angstroms) X	Y	Z
1	6	0	-0.082777	-0.049048	0.044261
2	1	0	0.632900	-0.281364	0.819381
3	1	0	0.429678	0.401216	-0.799549
4	1	0	-0.801520	0.672590	0.418563
5	6	0	-0.793227	-1.314194	-0.391771
6	8	0	-0.550223	-2.396621	0.093740
7	6	0	-1.829980	-1.124871	-1.480045
8	1	0	-2.281709	-2.075555	-1.723058
9	1	0	-1.367180	-0.705708	-2.367439
10	1	0	-2.597155	-0.432652	-1.148583

Transition State Optimization:

- iteratively modifying a molecular structure to arrive at a transition state
- the located structure should correspond to a saddle-point on the potential energy surface
- input structure must be reasonably close to the transition state
- require an accurate Hessian
- suitable for determining the structures and energies of transition states

This example:

- find the transition state for the reaction: $(CH_3)_2CO \rightarrow CH_3C(OH)CH_2$
- geometry provided in Z-matrix format
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Transition State Optimization - Input

```
# opt=(calcfc,ts,noeigen) hf/3-21g nosymm

Title Card Required

0 1
 C
 H    1   B1
 H    1   B2   2   A1
 H    1   B3   3   A2   2   D1
 C    1   B4   4   A3   3   D2
 O    5   B5   1   A4   4   D3
 C    5   B6   1   A5   6   D4
 H    7   B7   5   A6   1   D5
 H    7   B8   5   A7   1   D6
 H    7   B9   5   A8   1   D7

B1                 1.07000000
B2                 1.07000000
B3                 1.53523625
B4                 1.54000000
B5                 1.35030380
B6                 1.54000000
B7                 1.07000000
B8                 1.07000000
B9                 1.07000000
A1               109.47120255
A2               119.84000885
A3                70.47695871
A4               110.02650093
A5               120.00000000
A6               109.47122063
A7               109.47122063
A8               109.47122063
D1               144.21718948
D2              -109.72983768
D3                 0.00000000
D4              -180.00000000
D5               -20.26871844
D6              -140.26871844
D7                99.73128156
```

- keyword „opt“ requests a geometry optimization to a minimum energy structure
- option „ts“ requests a transition state optimization
- option „calcfc“ requests that the Hessian is calculated analytically at the first optimization step
- option „noeigen“ requests that the Hessian is not tested throughout the calculation. If testing is permitted, the calculation often fails because at some steps the Hessian may not have the correct number of imaginary frequencies.
- „hf/3-21G“ specifies level of theory for the calculation
- „nosymm“ tells the program to set the initial symmetry to C1
- geometry optimizations sometimes change the symmetry of the system, causing the calculation to fail
- the minimum energy structure may not have the same symmetry as the initial structure

Transition State Optimization – Output

- the output is the same as for a geometry optimization

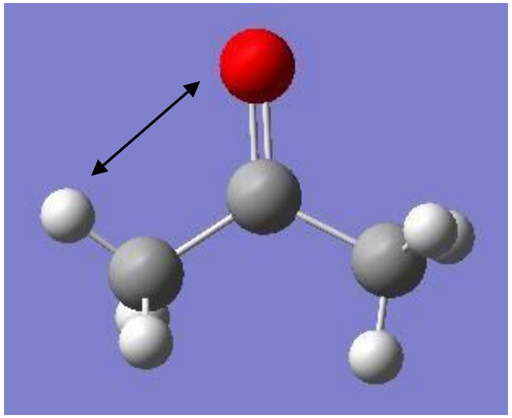
Relaxed Scan

Relaxed Scan Calculation:

- calculate the energies for a series of structures
- structures are based on predetermined changes to an initial structure, e.g. varying a bond length or changing an angle
- all non-scanned geometric variables are optimized, while scanned variables are held fixed
- provides a series of points on the potential energy surface
- useful for generating guess structures of transition states

This example:

- perform a relaxed scan of the O-H bond for $(CH_3)_2CO \rightarrow CH_3C(OH)CH_2$
- geometry provided in Z-matrix format $\rightarrow$ required by gaussian to do a scan
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Relaxed Scan - Input

```
# opt=(z-matrix) hf/3-21g nosymm

Title Card Required

0 1
 C
 H    1  B1
 H    1  B2   2  A1
 H    1  B3   3  A2   2  D1
 C    1  B4   4  A3   3  D2
 O    5  B5   1  A4   4  D3
 C    5  B6   1  A5   6  D4
 H    6  B7   5  A6   1  D5
 H    7  B8   5  A7   1  D6
 H    7  B9   5  A8   1  D7

B1      1.07000000
B2      1.07000000
B3      1.07000000
B4      1.54000000
B5      1.43000000
B6      1.54000000
B7      2.63480418   S   8   -0.2
B8      1.07000000
B9      1.07000000
A1    109.47120255
A2    109.47121829
A3    109.47121829
A4    120.00000000
A5    120.00000000
A6     54.53595458
A7    109.47122063
A8    109.47122063
D1    120.00000060
D2   -120.00003407
D3    180.00000000
D4   -180.00000000
D5   -173.71635255
D6     46.53624326
D7    -73.46375674
```

- keyword „opt" requests a geometry optimization to a minimum energy structure
  - option „z-matrix" requests that the optimization is performed using z-matrix coordinates
- „hf/3-21G" specifies level of theory for the calculation
- „nosymm" tells the program to set the initial symmetry to C1
  - geometry optimizations sometimes change the symmetry of the system, causing the calculation to fail
  - the minimum energy structure may not have the same symmetry as the initial structure

- „B7 2.63480418 S 8 -0.2" tells the program to decrease the value of variable B7 from its initial value in 8 steps of 0.2 Ang
- this will result in 9 calculations where the geometry is optimized except for bond length B7, which is held at the selected value
- all other variables will remain fixed at their original values
- more than one variable can be selected for scanning in a given input
- if two or more variables are scanned, the energy will be calculated for all possible combinations of the scanned variables

Relaxed Scan - Output

in a relaxed scan calculation we may want to determine:

- a series of energies and optimized structures as a function of the changed geometric variable

Summary of Optimized Potential Surface Scan

	1	2	3	4	5
EIGENVALUES --	-190.88668	-190.88677	-190.88266	-190.87354	-190.85791
B1	1.08022	1.08025	1.08026	1.08026	1.08024
B2	1.08500	1.08516	1.08522	1.08521	1.08512
B3	1.08500	1.08515	1.08522	1.08520	1.08513
B4	1.51779	1.51257	1.50767	1.50303	1.49851
B5	1.21150	1.21063	1.21056	1.21154	1.21386
B6	1.52267	1.50950	1.50165	1.49832	1.49911
B7	2.63480	2.43480	2.23480	2.03480	1.83480
B8	1.08555	1.08465	1.08371	1.08276	1.08190
B9	1.08554	1.08465	1.08371	1.08276	1.08191
A1	109.52267	109.55189	109.58587	109.62123	109.66843
A2	107.79292	107.73250	107.68830	107.65003	107.62435
A3	110.18684	110.07367	109.96887	109.88085	109.78381
A4	121.81005	123.06587	124.25795	125.34924	126.33353
A5	113.66799	116.31969	119.05926	121.90443	124.87340
A6	55.42482	59.18448	62.76280	66.08939	69.13664
A7	109.24334	110.87747	112.31507	113.54132	114.51955
A8	109.24404	110.88536	112.31592	113.54098	114.49103
D1	119.08993	119.11033	119.13761	119.18322	119.24161
D2	-120.27172	-120.02280	-119.82039	-119.63552	-119.46854
D3	239.39669	239.24941	239.22055	239.10746	239.14437
D4	-179.99933	-179.98113	-179.99823	-179.99740	-180.07406
D5	-179.99348	-180.01448	-180.00275	-179.99860	-180.02338
D6	58.54063	60.14984	61.70024	63.27129	64.68487
D7	-58.56014	-60.05762	-61.69068	-63.27244	-64.69538

step number in scan

energy at each step in Hartree

optimized z-matrix coordinates at each step

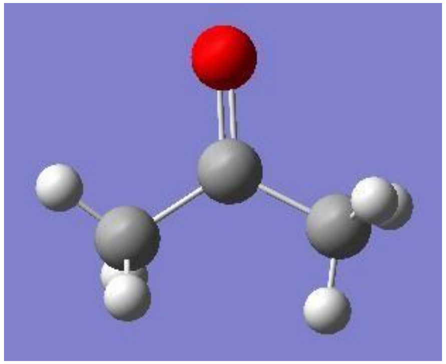
Frequency Calculation

Frequency Calculation:

- calculate the normal modes and associated vibrational frequencies for the input structure
- used to characterize stationary points as minima or transition states
- used to calculate zero-point vibrational energies
- used to calculate thermal corrections to the potential energy
- used to simulate IR/Raman spectra (future lecture)

This example:

- perform a frequency calculation of $(CH_3)_2CO$
- geometry provided in Z-matrix format
  $\rightarrow$ geometry obtained through a previous optimization
- calculation performed at hf/3-21G level of theory (more on this in future lectures)



Frequency Calculation - Input

```
# freq hf/3-21g

Title

0 1
 C
 H   1   B1
 H   1   B2   2   A1
 H   1   B3   3   A2   2   D1
 C   1   B4   4   A3   3   D2
 O   5   B5   1   A4   4   D3
 C   5   B6   1   A5   6   D4
 H   7   B7   5   A6   1   D5
 H   7   B8   5   A7   1   D6
 H   7   B9   5   A8   1   D7

   B1       1.07000000
   B2       1.07000000
   B3       1.07000000
   B4       1.54000000
   B5       1.43000000
   B6       1.54000000
   B7       1.07000000
   B8       1.07000000
   B9       1.07000000
   A1     109.47120255
   A2     109.47121829
   A3     109.47121829
   A4     120.00000000
   A5     120.00000000
   A6     109.47122063
   A7     109.47122063
   A8     109.47122063
   D1     120.00000060
   D2    -120.00003407
   D3     180.00000000
   D4    -180.00000000
   D5     166.53624326
   D6      46.53624326
   D7     -73.46375674
```

- keyword „freq" requests a frequency calculation
- „hf/3-21G" specifies level of theory for the calculation

- coordinates in Z-matrix format, but frequency calculations can also be performed with cartesian coordinates
- structure must correspond to a stationary point
  - recall, frequency calculations in harmonic approximation are only valid at stationary points

Frequency Calculation - Output

in a frequency calculation we may want to determine:

- normal modes and vibrational frequencies

```
Harmonic frequencies (cm**-1), IR intensities (KM/Mole), Raman scattering
activities (A**4/AMU), depolarization ratios for plane and unpolarized
incident light, reduced masses (AMU), force constants (mDyne/A),
and normal coordinates:
                          1                       2                       3
                          A                       A                       A
Frequencies --     85.8357                160.7480                405.9040
Red. masses --      1.0197                  1.1162                  2.2117
Frc consts  --      0.0044                  0.0170                  0.2147
IR Inten    --      0.0000                  0.0462                  1.1897
Raman Activ --      0.0519                  0.0007                  0.2617
Depolar (P) --      0.7500                  0.7493                  0.2558
Depolar (U) --      0.8571                  0.8567                  0.4074
Atom AN      X      Y      Z        X      Y      Z        X      Y      Z
  1   6    0.00   0.00   0.02     0.00   0.00   0.04     0.16   0.10   0.00
  2   1    0.00   0.00  -0.35     0.00   0.00   0.41    -0.06   0.39   0.00
  3   1    0.16   0.32   0.25    -0.17  -0.31  -0.19     0.36   0.12   0.01
  4   1   -0.16  -0.32   0.25     0.17   0.31  -0.19     0.36   0.12  -0.01
  5   6    0.00   0.00   0.00     0.00   0.00   0.02     0.00  -0.13   0.00
  6   8    0.00   0.00   0.00     0.00   0.00  -0.07     0.00  -0.13   0.00
  7   6    0.00   0.00  -0.02     0.00   0.00   0.04    -0.16   0.10   0.00
  8   1    0.00   0.00   0.35     0.00   0.00   0.41     0.06   0.39   0.00
  9   1    0.16  -0.32  -0.25     0.17  -0.31  -0.19    -0.36   0.12   0.01
 10   1   -0.16   0.32  -0.25    -0.17   0.31  -0.19    -0.36   0.12  -0.01
```

mode

frequencies in cm^{-1}

normal mode displacements

Frequency Calculation - Output

in a frequency calculation we may want to determine:

- normal modes and vibrational frequencies
  - zero-point vibrational energies

```
Zero-point vibrational energy     236075.8 (Joules/Mol)
                                  56.42347 (Kcal/Mol)
```

- thermal corrections to the potential energy

```
Zero-point correction=                           0.089917 (Hartree/Particle)
Thermal correction to Energy=                    0.095029
Thermal correction to Enthalpy=                  0.095973
Thermal correction to Gibbs Free Energy=         0.062198
Sum of electronic and zero-point Energies=           -190.797305
Sum of electronic and thermal Energies=              -190.792192
Sum of electronic and thermal Enthalpies=            -190.791248
Sum of electronic and thermal Free Energies=         -190.825023
```